Here, replace *idVendor* with the ID of your phone manufacturer, which can usually be found in the manufacturers list [see Reference 4]. As the root, restart the *udev* subsystem with */etc/init.d/boot.udev*, and then disconnect and re-plug the USB cord.

```
[1001646.344032] hub 2-0:1.0: unable to enumerate USB device on
port 1
[1001647.084044] usb 2-1: new high speed USB device using ehci_
hcd and address 13
[1001647.216857] usb 2-1: New USB device found, idVendor=0bb4,
idProduct=0c03
[1001647.216861] usb 2-1: New USB device strings: Mfr=1,
Product=2, SerialNumber=3
[1001647.216864] usb 2-1: Product: MT65xx Android Phone
[1001647.216866] usb 2-1: Manufacturer: MediaTek
[1001647.216868] usb 2-1: SerialNumber: MT-
S9NPaAMhQdBBD1gBEBDfAxvA
[1001647.217399] scsi36 : usb-storage 2-1:1.0
[1001648.235664] scsi 36:0:0:0: Direct-Access     MediaTek MT65xx
MS       0100 PQ: 0 ANSI: 2
[1001648.235825] sd 36:0:0:0: Attached scsi generic sg2 type 0
[1001648.239143] sd 36:0:0:0: [sdb] 3964677 512-byte logical
blocks: (2.02 GB/1.88 GiB)
[1001648.240137] sd 36:0:0:0: [sdb] Write Protect is off
[1001648.240140] sd 36:0:0:0: [sdb] Mode Sense: 03 00 00 00
[1001648.240143] sd 36:0:0:0: [sdb] Assuming drive cache: write
through
[1001648.242886] sd 36:0:0:0: [sdb] Assuming drive cache: write
through
[1001648.242893]  sdb:
[1001648.247761] sd 36:0:0:0: [sdb] Assuming drive cache: write
through
[1001648.247764] sd 36:0:0:0: [sdb] Attached SCSI removable disk
```

Why exactly were these parameters used? Just because my smartphone has this manufacturer ID, although this is not a device from HTC at all!

```
$ lsusb | grep 0bb4
Bus 002 Device 015: ID 0bb4:0c03 High Tech Computer Corp.
```

ADB access

Run *adb* now, and enjoy your access to the Android prompt on the phone!

```
$ ./platform-tools/adb shell
$ cat /proc/cpuinfo
Processor       : ARM926EJ-S rev 5 (v5l)
BogoMIPS        : 207.66
Features        : swp half thumb fastmult edsp java
CPU implementer : 0x41
CPU architecture: 5TEJ
CPU variant     : 0x0
CPU part        : 0x926
CPU revision    : 5
Hardware        : MT6516 E1K
Revision        : 659e8b01
Serial          : 0000000000000000
```

You can see that magical BogoMIPS number—frankly, I don't know how well it reflects the actual frequency of the CPU, because MediaTek itself declares the CPU to be running at 416 MHz. Even with ordinary user rights, you can do many interesting things, such as scanning the Wi-Fi network:

```
$ iwlist wlan0 scan
wlan0     Scan completed :
          Cell 01 - Address: 00:15:D1:4B:E6:F7
                    ESSID:"my_net"
                    Frequency:2.412 GHz (Channel 1)
                    Mode:Managed
                    Signal level=-65 dBm
                    Encryption key:off
                    Bit Rates:54 Mb/s
                    Extra:Rates (Mb/s): 1 2 5.5 11 6 9 12 18 24
36 48 54
          Cell 02 - Address: F1:7D:68:48:C2:A0
                    ESSID:"dd-wrt"
                    Frequency:2.437 GHz (Channel 6)
                    Mode:Managed
                    Signal level=-68 dBm
                    Encryption key:off
                    Bit Rates:54 Mb/s
                    Extra:Rates (Mb/s): 1 2 5.5 11 6 9 12 18 24
36 48 54
```

You can find the IP address that DHCP assigned to the smartphone when it joined the wireless network with *ifconfig wlan0*, display the phone's installed RAM with *cat /proc/meminfo*, or run *mount* to view mounted block devices, besides other commands you'd expect. The basic Android system commands are located in */system/bin* and */system/sbin*. You can list them and see that */system/bin* has almost the same commands as those in the */bin* directory on a Linux desktop installation. However, there are several special features, like the service command, which lets you display the full list of services by running *service list*.

All Android OS parameters are available via the *getprop* command. For instance, a question that tortured me for months was exactly what type of touch-screen the device had—resistive or capacitive? The *getprop* output has *[ro.kernel.touchpanel.type]:[capacitive]*, but I doubt it, because I can use my pencil on the virtual keyboard, and it works. This is impossible with a real capacitive screen, like that used in Apple products. So, the question is still open :-).